Thermal Emission from Geometrically Thin Accretion Disks: The Role of Black Hole Mass and Accretion Rate

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Abstract

We investigate how black hole mass and accretion rate shape the thermal continuum of a geometrically thin, optically thick accretion disk. The disk is modelled as a set of concentric rings, each radiating as a blackbody at the local effective temperature of the steady Newtonian thin-disk solution with a zero-torque inner boundary at the Schwarzschild innermost stable circular orbit, as implemented in the published `agnpy` package, with the Planck function and physical constants taken from `astropy`. The emission of all rings is summed numerically to obtain the specific luminosity L_ν and the flux F_ν received by an observer at a given distance and inclination. The numerical scheme reproduces the analytic bolometric luminosity to a fractional accuracy of 5×10^{-6} and converges as N^{-2} in the number of rings. At a fixed Eddington ratio $L/L_{\rm Edd} = 0.1$, a $10\,M_\odot$ black hole has a maximum disk temperature of $T_{\rm max} = 4.4 \times 10^6$ K and a spectral peak at ≈ 0.94 keV (soft X-rays), while a $10^8\,M_\odot$ black hole has $T_{\rm max} = 7.9 \times 10^4$ K and peaks at ≈ 740 Å (far ultraviolet). The two peaks differ by exactly $(10^7)^{1/4} \approx 56$, reflecting the scaling $T_{\rm max} \propto (\dot{M}/M^2)^{1/4}$. The bolometric luminosity depends only on the accretion rate, $L = \dot{M}c^2/12$, so the luminosity and the spectral peak together determine both $\dot{M}$ and M. We compare the model with real observations. In SDSS quasars it matches the average far-ultraviolet continuum slope but is too blue in the near-ultraviolet and optical, where 86–88% of $\sim 10^5$ quasars are redder than the disk predicted for their own mass and Eddington ratio. Fitted to 48 thermal-state RXTE spectra of the black hole binary LMC X-3, with its dynamically measured mass, it gives good fits (median $\chi^2_\nu = 0.79$) with a nearly constant spectral hardening factor $f_{\rm col} = 1.81 \pm 0.09$, close to the theoretical value of ≈ 1.7. We discuss the limitations imposed by neglecting general-relativistic effects, non-blackbody radiative transfer and additional emission components.

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1 Introduction

Accretion onto black holes is among the most efficient ways of converting rest-mass energy into radiation. Black holes span more than eight orders of magnitude in mass, from stellar-mass objects of $\sim$ 5–20 $M_\odot$ in X-ray binaries to supermassive black holes of 10^6–10^{10} $M_\odot$ in active galactic nuclei (AGN). Despite this range, the same basic physics is thought to operate in both: gas with angular momentum settles into a disk, viscous stresses transport angular momentum outward and allow the gas to spiral inward, and the gravitational energy released is radiated from the disk surface.

The standard model for this process is the geometrically thin, optically thick disk of Shakura and Sunyaev [1], reviewed by Pringle [2] and Frank et al. [3]. Because the disk is optically thick, each annulus radiates approximately as a blackbody, and because the temperature falls with radius, the total spectrum is a superposition of blackbodies with a range of temperatures. This "multi-colour disk" spectrum is used to describe the thermal component of X-ray binaries in their soft state [4, 5] and the "big blue bump" seen in the optical–ultraviolet spectra of quasars [6, 7].

In this project we implement this model numerically and ask three questions:

1. How do the black hole mass M and accretion rate $\dot{M}$ set the disk temperature, the spectral peak and the total luminosity?

2. What flux would an observer at a given distance and inclination measure?

3. Why does the same physical model put the emission of stellar-mass black holes in the X-ray band and that of supermassive black holes in the ultraviolet?

4. How well does the model describe real observations of quasars and of a stellar-mass black hole?

Section 2 summarises the physical model, Section 3 describes the numerical method and its validation, Section 4 presents the results, Section 5 compares the model with observations, and Section 6 connects them to observables and discusses the limitations of the model.

2 Physical model

2.1 Assumptions

We consider a steady, axisymmetric disk around a non-rotating black hole of mass M accreting at a constant rate $\dot{M}$. The disk is assumed to be

- *geometrically thin*, $H/r \ll 1$, so that it can be treated as a flat surface and radial pressure support is negligible;
- *optically thick*, so that each surface element radiates as a blackbody at its local effective temperature;
- *Keplerian and Newtonian*, with the inner edge at the innermost stable circular orbit (ISCO) of a Schwarzschild black hole, $r_{\rm in} = 6r_g$, where $r_g = GM/c^2$ is the gravitational radius.

Unless stated otherwise the outer radius is $r_{\rm out} = 10^5 r_g$.

2.2 Radial temperature profile

Conservation of mass and angular momentum in a steady thin disk, together with the assumption that the viscous torque vanishes at $r_{\rm in}$, gives the rate of energy dissipation per unit area of each face [1, 3]:

$$D(r) = \frac{3GM\dot{M}}{8\pi r^3}\left[1 - \left(\frac{r_{\rm in}}{r}\right)^{1/2}\right]. \tag{1}$$

In local thermal equilibrium this energy is radiated as a blackbody, so $\sigma T^4(r) = D(r)$ and

$$T(r) = T_*\left(\frac{r_{\rm in}}{r}\right)^{3/4}\left[1 - \left(\frac{r_{\rm in}}{r}\right)^{1/2}\right]^{1/4}, \qquad T_* = \left(\frac{3GM\dot{M}}{8\pi\sigma r_{\rm in}^3}\right)^{1/4}. \tag{2}$$

Far from the inner edge $T \propto r^{-3/4}$. The temperature vanishes at $r_{\rm in}$ and reaches its maximum at $r = \frac{49}{36} r_{\rm in}$, where

$$T_{\rm max} = \left[\frac{(36/49)^3}{7}\right]^{1/4} T_* \approx 0.488\, T_*. \tag{3}$$

Because $r_{\rm in} \propto M$, the temperature scale obeys

$$T_{\rm max} \propto \left(\frac{\dot{M}}{M^2}\right)^{1/4}. \tag{4}$$

2.3 Luminosity and the Eddington limit

Integrating $2D(r)$ over both faces from $r_{\rm in}$ to $r_{\rm out}$ gives the exact bolometric luminosity. With $X = r_{\rm out}/r_{\rm in}$,

$$L = \frac{3GM\dot{M}}{2r_{\rm in}}\left[\left(1 - \frac{1}{X}\right) - \frac{2}{3}\left(1 - X^{-3/2}\right)\right] \xrightarrow{X\to\infty} \frac{GM\dot{M}}{2r_{\rm in}} = \frac{\dot{M}c^2}{12}. \tag{5}$$

The radiative efficiency of this Newtonian model is therefore $\eta = 1/12$, and the luminosity depends on $\dot{M}$ alone. We parametrise the accretion rate through the Eddington ratio $\lambda = L/L_{\rm Edd}$, where

$$L_{\rm Edd} = \frac{4\pi GMm_p c}{\sigma_T} \approx 1.26 \times 10^{38}\,{\rm erg\,s^{-1}}\left(\frac{M}{M_\odot}\right) \tag{6}$$

is the luminosity at which radiation pressure on electrons balances gravity for ionised hydrogen. The accretion rate is then $\dot{M} = \lambda L_{\rm Edd}/(\eta c^2)$; for a $10\,M_\odot$ black hole at $\lambda = 1$ this is $\dot{M} = 1.68 \times 10^{19}\,{\rm g\,s^{-1}} = 2.7 \times 10^{-7}\,M_\odot\,{\rm yr^{-1}}$. At fixed λ, $\dot{M} \propto M$, so Eq. (4) gives $T_{\rm max} \propto M^{-1/4}$: *more massive black holes have cooler disks.*

2.4 Spectrum

Each annulus of area $dA = 2\pi r\, dr$ per face emits isotropically with specific intensity $B_\nu(T)$, the Planck function [8]

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT} - 1}. \tag{7}$$

The specific luminosity emitted from both faces is

$$L_\nu = 2\int_{r_{\rm in}}^{r_{\rm out}} \pi B_\nu\big(T(r)\big)\, 2\pi r\, dr. \tag{8}$$

An observer at distance D who views the disk at inclination i (with $i = 0$ face-on) sees a projected area $dA\cos i$, so the received flux density is

$$F_\nu = \frac{\cos i}{D^2}\int_{r_{\rm in}}^{r_{\rm out}} B_\nu\big(T(r)\big)\, 2\pi r\, dr = \frac{\cos i}{2\pi D^2}\, L_\nu. \tag{9}$$

The spectrum has three regimes. For $h\nu \ll kT(r_{\rm out})$ every annulus is in the Rayleigh–Jeans limit and $L_\nu \propto \nu^2$. For $h\nu \gg kT_{\rm max}$ the spectrum is cut off exponentially, as in the Wien tail. In between, the substitution $x = h\nu/kT(r)$ with $T \propto r^{-3/4}$ gives $r \propto x^{4/3}\nu^{-4/3}$, and Eq. (8) becomes

$$L_\nu \propto \nu^3 \int \frac{r\, dr}{e^x - 1} \propto \nu^3\, \nu^{-8/3} \int \frac{x^{5/3}\, dx}{e^x - 1} \propto \nu^{1/3}, \tag{10}$$

the characteristic signature of a multi-temperature disk.

3 Numerical method

3.1 Model implementation and software

Rather than re-deriving the disk model, we build on established, published software:

- **Disk model.** The temperature profile, Eq. (2), is taken from the `SSDisk` class of `agnpy` [9], an open-source, peer-reviewed Python package for the radiative processes of active galactic nuclei. `SSDisk` implements the Shakura and Sunyaev [1] disk with a zero-torque inner boundary, following Dermer and Menon [10]. agnpy describes a disk by $(M, L_{\rm disk}, \eta, r_{\rm in}, r_{\rm out})$; we pass $L_{\rm disk} = \eta\dot{M}c^2$ with $\eta = 1/12$, so agnpy's accretion rate equals ours exactly.
- **Planck function and constants.** $B_\nu(T)$ is evaluated with the `BlackBody` model of `astropy` [11], and all physical constants come from `astropy.constants` (CODATA 2018 and IAU 2015 values).

What this project adds is the integration over the disk (Section 3.2). agnpy can also compute the disk spectrum itself (`SSDisk.evaluate_multi_T_bb_sed`), using the same flux formula as Eq. (9), but it integrates over 100 *linearly* spaced radii. That is accurate for compact disks, the case agnpy was designed for, but for $r_{\rm out} = 10^5 r_g$ the grid spacing is $\sim 10^3 r_g$ and the hot inner disk is not resolved. We therefore use agnpy's temperature profile with our own logarithmically spaced ring sum, and verify the two against each other in Section 3.4.

The same Shakura–Sunyaev spectrum is also standard in X-ray astronomy: it is available in XSPEC [12] as `ezdiskbb` [13], which uses the same zero-torque profile, and in the simpler `diskbb` [4], which uses a pure $T \propto r^{-3/4}$ law.

3.2 Ring discretisation

The code (`accretion_disk/disk.py`) works in CGS units throughout. The disk is divided into N rings whose edges $r_0 = r_{\rm in} < r_1 < \cdots < r_N = r_{\rm out}$ are logarithmically spaced, so every decade in radius receives equal resolution. Ring k is assigned the temperature $T_k = T(\bar{r}_k)$ at its geometric-mean radius $\bar{r}_k = \sqrt{r_k r_{k+1}}$ and the one-face area $A_k = \pi(r_{k+1}^2 - r_k^2)$. Equations (8) and (9) become sums over rings,

$$L_\nu = 2\pi \sum_{k=0}^{N-1} B_\nu(T_k)\, A_k, \qquad F_\nu = \frac{\cos i}{D^2} \sum_{k=0}^{N-1} B_\nu(T_k)\, A_k, \tag{11}$$

which are evaluated for all frequencies at once as a matrix–vector product. The default is $N = 2000$. The Planck function is set to exactly zero for $h\nu/kT > 600$, where it is below 10^{-255} of its peak, to avoid floating-point overflow in the Wien tail.

Figure 1 illustrates the method: grouping the 2000 rings into eight zones shows that each zone contributes a nearly blackbody spectrum, while their sum is the broad multi-temperature disk spectrum.

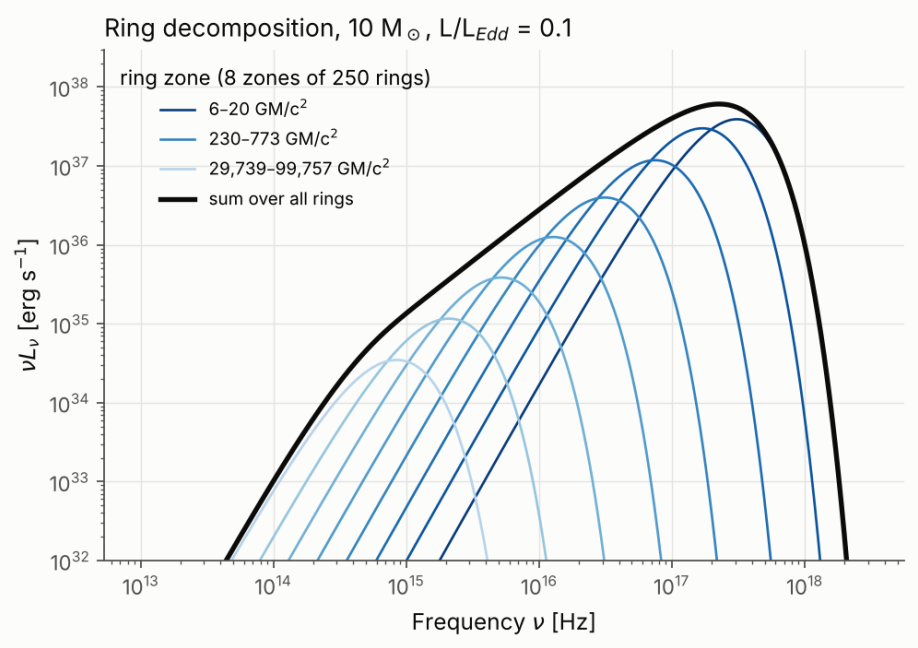


Figure 1: Decomposition of the disk spectrum of a $10\,M_\odot$ black hole at $\lambda = 0.1$. Coloured curves show the summed emission of eight zones of 250 rings each, from the hot inner zone (dark) to the cool outer zone (light); the black curve is the total. Each zone is close to a blackbody, but the total is much broader.

3.3 Derived quantities

The bolometric luminosity is computed directly from the rings as $L = 2\sigma \sum_k T_k^4 A_k$. The peak frequency $\nu_{\rm peak}$ of νL_ν is found on a logarithmic grid of 4000 frequencies spanning 10^{-5} to 40 times $kT_{\rm max}/h$; the grid maximum is refined by fitting a parabola to the three neighbouring points in $\log\nu$–$\log(\nu L_\nu)$ space. Scaling the grid with $T_{\rm max}$ gives the same relative resolution for every mass and accretion rate.

3.4 Validation

The implementation is checked by an automated test suite (`tests/test_disk.py`, 22 tests). Table 1 summarises the main checks against analytic results and against agnpy.

Table 1: Validation of the numerical model against analytic results and against agnpy.

Check	Result
$\pi \int B_\nu \, d\nu = \sigma T^4$	agrees to $< 10^{-4}$
Ring sum of σT^4 vs. Eq. (5)	5.4×10^{-6} ($N = 2000$)
$\int L_\nu \, d\nu$ vs. Eq. (5)	9.5×10^{-6}
Location and value of $T_{\max}$	$r = \frac{49}{36} r_{\rm in}$, $T_{\max} = 0.488\, T_*$
Low-frequency slope	$L_\nu \propto \nu^{2.00}$
Intermediate slope ($r_{\rm out} = 10^8 r_g$)	$L_\nu \propto \nu^{0.33}$
Mass scaling at fixed λ	$\nu_{\rm peak} \propto M^{-0.2500}$
Accretion-rate scaling at fixed M	$\nu_{\rm peak} \propto \dot{M}^{0.2500}$, $L \propto \dot{M}^{1.000}$
Observer geometry	$F_\nu \propto \cos i / D^2$, face-on $F_\nu = L_\nu / 2\pi D^2$
$T(r)$ vs. closed form, Eq. (2)	identical to $< 10^{-10}$
νF_ν vs. agnpy SED ($r_{\rm out} = 50 r_g$)	agrees to $< 0.3\,\%$

Figure 2 shows the convergence of the ring sum. The error in the bolometric luminosity falls as N^{-2}, as expected for a midpoint-type rule; about 100 rings give 0.2 % accuracy, and the default of 2000 rings gives errors well below 10^{-5}, negligible compared with the physical uncertainties of the model.

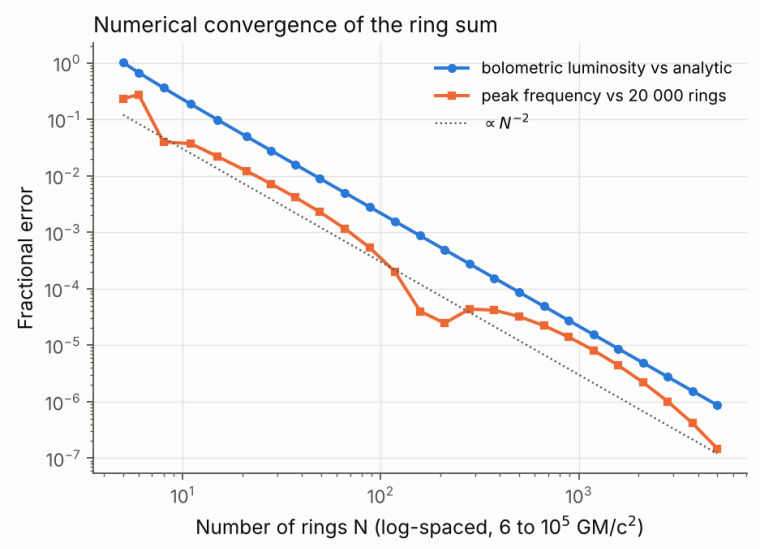


Figure 2: Fractional error of the bolometric luminosity (relative to Eq. 5) and of the peak frequency (relative to a 20 000-ring calculation) as a function of the number of rings N.

Figure 3 compares our ring sum with agnpy's own spectrum calculation. For compact disks ($r_{\rm out} \leq 50 r_g$), where agnpy's 100-point linear grid is adequate, the two agree to 0.1 % at all frequencies that carry significant flux. This confirms that our code computes the same Shakura–Sunyaev spectrum as the published package. As $r_{\rm out}$ grows, agnpy's grid becomes too coarse to resolve the inner disk: at $r_{\rm out} = 10^3 r_g$ it underestimates the flux at $h\nu = 10\, kT_{\max}$ by a factor of five, and at $r_{\rm out} = 10^5 r_g$ it misses the spectral peak entirely.

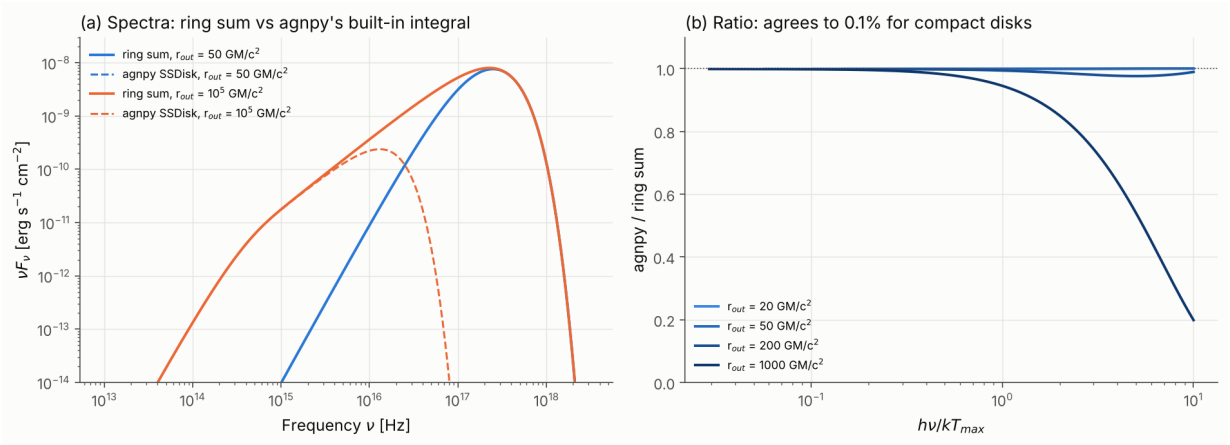


Figure 3: (a) νF_ν of a $10\,M_\odot$ disk at $\lambda = 0.1$ (8 kpc, $i = 60°$) from our ring sum (solid) and from agnpy's built-in integration (dashed), for $r_{\rm out} = 50 r_g$ (the curves overlap) and $r_{\rm out} = 10^5 r_g$. (b) Ratio of agnpy's result to the ring sum for several disk sizes.

4 Results

We adopt two representative systems, both accreting at $\lambda = 0.1$:

- a **stellar-mass black hole** of $10\,M_\odot$ at $D = 8\,$kpc viewed at $i = 60°$, typical of a Galactic X-ray binary;
- a **supermassive black hole** of $10^8\,M_\odot$ at $D = 100\,$Mpc viewed at $i = 30°$, typical of a Seyfert nucleus or quasar.

Their properties at three Eddington ratios are listed in Table 2.

Table 2: Disk properties of the representative systems. $\nu_{\rm peak}$ and $E_{\rm peak} = h\nu_{\rm peak}$ refer to the peak of νL_ν; $F_{\rm bol}$ is the bolometric flux at the stated distance and inclination ($D = 8\,$kpc, $i = 60°$ for $10\,M_\odot$; $D = 100\,$Mpc, $i = 30°$ for $10^8\,M_\odot$).

M $[M_\odot]$	$L/L_{\rm Edd}$	$\dot{M}$ $[M_\odot\,{\rm yr}^{-1}]$	$L_{\rm bol}$ $[{\rm erg\,s}^{-1}]$	$T_{\rm max}$ [K]	$\nu_{\rm peak}$ [Hz]	$E_{\rm peak}$ [keV]	$F_{\rm bol}$ $[{\rm erg\,s}^{-1}\,{\rm cm}^{-2}]$
10	0.01	2.7×10^{-9}	1.26×10^{37}	2.49×10^{6}	1.28×10^{17}	0.53	1.6×10^{-9}
10	0.1	2.7×10^{-8}	1.26×10^{38}	4.42×10^{6}	2.27×10^{17}	0.94	1.6×10^{-8}
10	1	2.7×10^{-7}	1.26×10^{39}	7.86×10^{6}	4.04×10^{17}	1.67	1.6×10^{-7}
10^8	0.01	2.7×10^{-2}	1.26×10^{44}	4.42×10^{4}	2.27×10^{15}	0.0094	1.8×10^{-10}
10^8	0.1	2.7×10^{-1}	1.26×10^{45}	7.86×10^{4}	4.04×10^{15}	0.0167	1.8×10^{-9}
10^8	1	2.7	1.26×10^{46}	1.40×10^{5}	7.18×10^{15}	0.0297	1.8×10^{-8}

4.1 Temperature structure

Figure 4 shows the temperature profiles of the two systems. Expressed in units of r_g, the profiles have identical shapes: a steep rise from zero at the ISCO, a maximum at $8.2\,r_g$, and a $T \propto r^{-3/4}$ decline. Only the normalisation differs. The inner disk of the $10\,M_\odot$ black hole reaches $4.4\times10^6\,$K, while that of the $10^8\,M_\odot$ black hole reaches only $7.9\times10^4\,$K, even though it radiates 10^7 times more power. The ratio of the two temperatures is exactly $(10^7)^{1/4} = 56.2$, as required by Eq. (4). The supermassive disk can be cooler because its emitting area, $\propto M^2$, is 10^{14} times larger. The zero-torque boundary condition lowers the temperature at the maximum by a factor of $7^{1/4} \approx 1.6$ relative to the pure power law (dotted lines), and drives it to zero at $r_{\rm in}$.

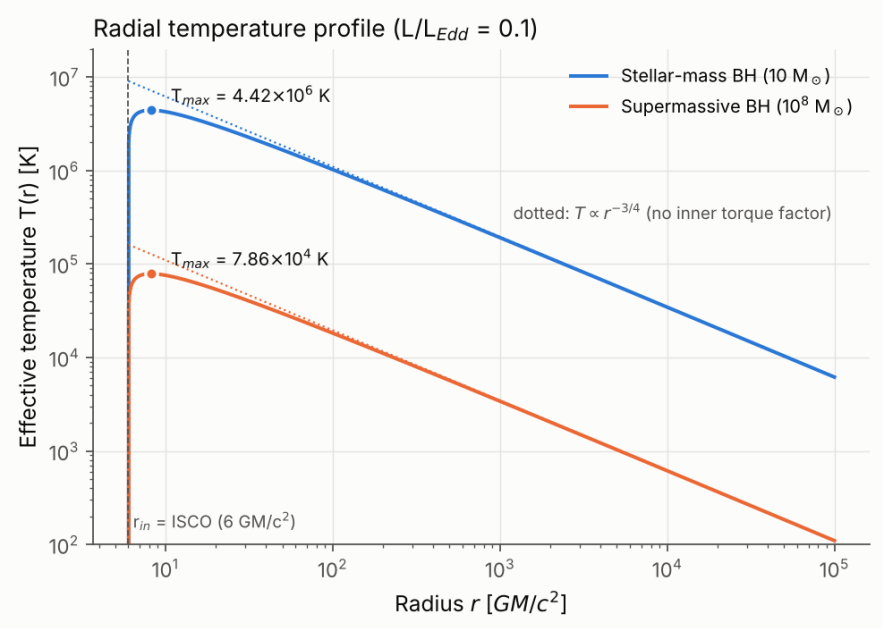


Figure 4: Effective temperature as a function of radius for a $10\,M_\odot$ and a $10^8\,M_\odot$ black hole at $\lambda = 0.1$. Dots mark $T_{\max}$ at $r = \frac{49}{36} r_{\rm in}$; dotted lines show the $T \propto r^{-3/4}$ power law without the inner-boundary factor.

4.2 Stellar-mass versus supermassive black holes

Figure 5 compares the spectra. The $10\,M_\odot$ disk peaks at $\nu_{\rm peak} = 2.27 \times 10^{17}$ Hz ($E \approx 0.94$ keV), in the soft X-ray band, which matches the thermal disk component observed in black hole X-ray binaries in their high/soft state [5]. The $10^8\,M_\odot$ disk peaks at $\nu_{\rm peak} = 4.04 \times 10^{15}$ Hz ($\lambda \approx 740$ Å), in the far ultraviolet, consistent with the big blue bump in quasar spectra [6, 7].

The spectral *shape* is the same in both cases when frequency is expressed in units of $kT_{\max}/h$: we find $h\nu_{\rm peak} = 2.466\,kT_{\max}$ for the peak of νL_ν and $h\nu = 0.80\,kT_{\max}$ for the peak of L_ν, independent of M and $\dot{M}$. The single model therefore produces characteristic frequencies that differ by a factor of 56 simply because $T_{\max} \propto M^{-1/4}$ at fixed Eddington ratio. Both spectra show the three regimes of Section 2: a Rayleigh–Jeans ν^2 tail below the temperature of the outer edge ($T(r_{\rm out}) \approx 6200$ K for the stellar disk and ≈ 110 K for the supermassive disk), a $\nu^{1/3}$ section ($\nu L_\nu \propto \nu^{4/3}$) at intermediate frequencies, and a Wien cutoff above $\nu_{\rm peak}$.

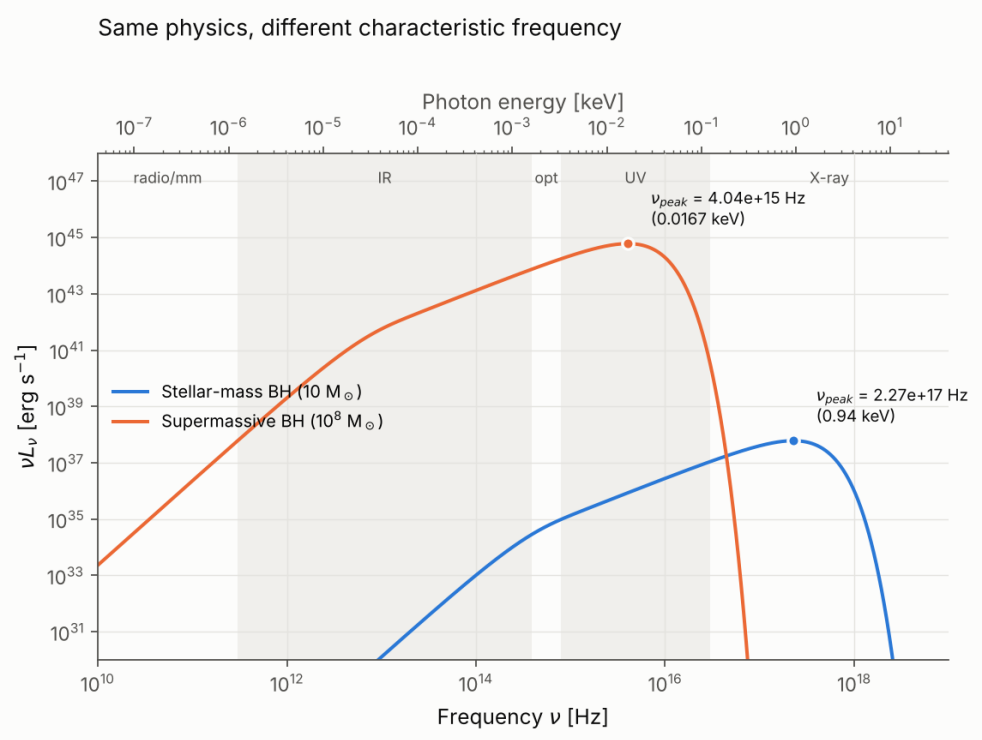


Figure 5: Spectral energy distributions νL_ν of the $10\,M_\odot$ and $10^8\,M_\odot$ disks at $\lambda = 0.1$. Shaded bands mark the IR and UV regions; the top axis gives the photon energy.

4.3 Dependence on black hole mass

Figure 6 shows a sequence of masses from 10 to $10^9\,M_\odot$ at fixed $\lambda = 0.1$. Increasing the mass raises the luminosity in proportion ($L \propto \dot{M} \propto M$) and moves the peak to lower frequency, so the peaks lie along the locus $\nu L_\nu \propto \nu_{\rm peak}^{-4}$ (dashed line). A power-law fit to the peak frequencies gives a slope of -0.2500, in exact agreement with $\nu_{\rm peak} \propto M^{-1/4}$. Over this mass range the peak moves from the X-ray band, through the extreme ultraviolet, into the far ultraviolet.

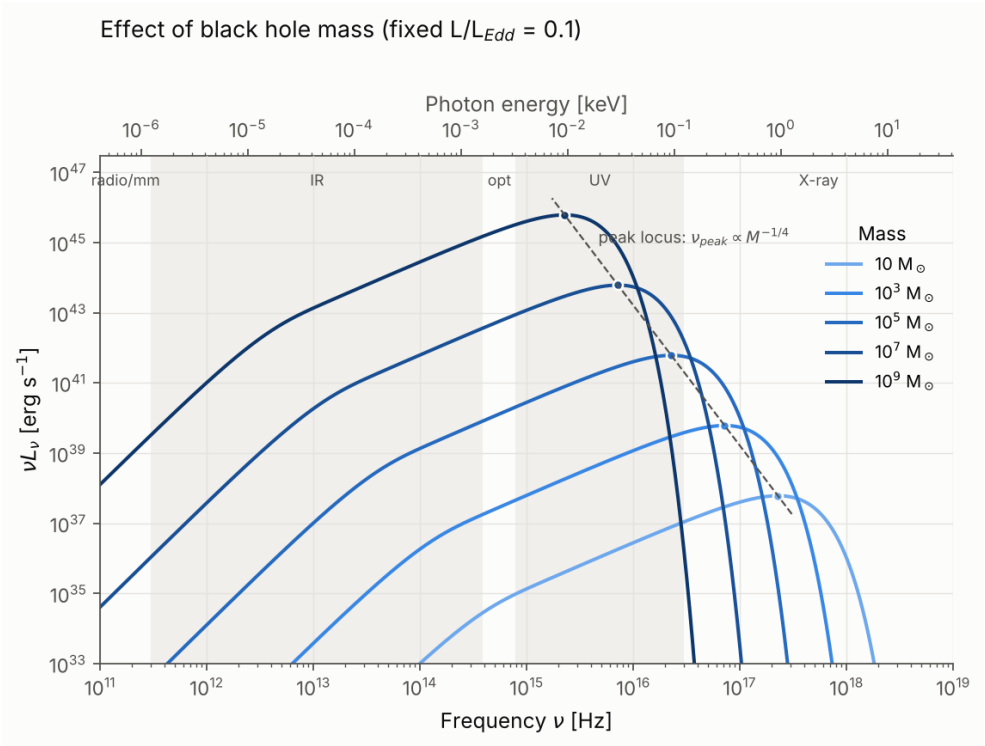


Figure 6: Disk spectra for black hole masses of 10, 10^3, 10^5, 10^7 and $10^9\,M_\odot$ at $\lambda = 0.1$. Dots mark the peak of each spectrum; the dashed line traces the peak locus.

4.4 Dependence on accretion rate

Figure 7 shows the effect of varying the accretion rate from $\lambda = 0.01$ to 1 at fixed mass. The luminosity increases in proportion to $\dot{M}$, but the peak moves only as $\dot{M}^{1/4}$: a hundredfold increase in accretion rate shifts $\nu_{\rm peak}$ by a factor of only $100^{1/4} \approx 3.2$ (from 0.53 to 1.67 keV for the stellar-mass disk). In the $\nu^{1/3}$ regime the spectra at different $\dot{M}$ are parallel, with $L_\nu \propto \dot{M}^{2/3}$ at fixed frequency.

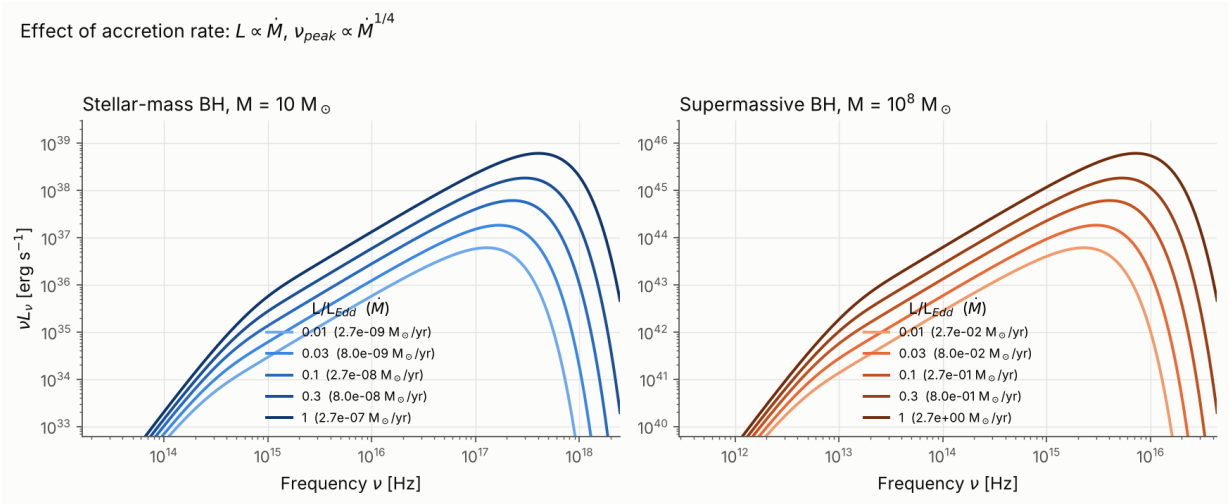


Figure 7: Disk spectra for Eddington ratios $\lambda = 0.01$–1 for the $10\,M_\odot$ (left) and $10^8\,M_\odot$ (right) black holes. The legend gives the corresponding accretion rate.

4.5 Scaling relations

Figure 8 summarises the scaling laws. At fixed Eddington ratio $\nu_{\rm peak} \propto M^{-1/4}$ (panel a); at fixed mass $\nu_{\rm peak} \propto \dot{M}^{1/4}$ (panel b); and the bolometric luminosity lies on the single relation $L = \dot{M}c^2/12$ for every mass (panel c). Combining these, the model results can be written compactly as

$$T_{\rm max} \approx 4.4 \times 10^6\,{\rm K} \left(\frac{M}{10\,M_\odot}\right)^{-1/4} \left(\frac{\lambda}{0.1}\right)^{1/4}, \tag{12}$$

$$\nu_{\rm peak} \approx 2.3 \times 10^{17}\,{\rm Hz} \left(\frac{M}{10\,M_\odot}\right)^{-1/4} \left(\frac{\lambda}{0.1}\right)^{1/4}. \tag{13}$$

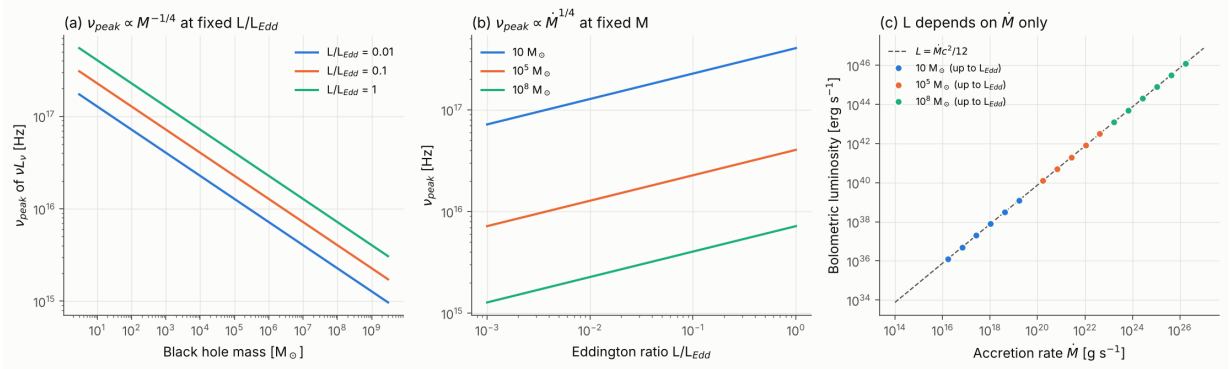


Figure 8: (a) Peak frequency versus black hole mass at three Eddington ratios. (b) Peak frequency versus Eddington ratio at three masses. (c) Bolometric luminosity versus accretion rate; points for all masses lie on $L = \dot{M}c^2/12$.

4.6 Observed flux: distance and inclination

Figure 9 shows the flux received by observers of the two fiducial systems at several inclinations. At $\lambda = 0.1$ the X-ray binary at 8 kpc and $i = 60°$ has a bolometric flux of $1.6 \times 10^{-8}\,{\rm erg\,s^{-1}\,cm^{-2}}$, typical of a bright Galactic X-ray source. The AGN at 100 Mpc and $i = 30°$ has $1.8 \times 10^{-9}\,{\rm erg\,s^{-1}\,cm^{-2}}$, its far greater luminosity being largely offset by its 1.25×10^4 times greater distance.

In this non-relativistic model inclination affects only the normalisation through $\cos i$; the spectral shape is unchanged. Going from face-on to $i = 85°$ reduces the flux by a factor of 11. An observer who assumes isotropic emission and infers $L_{\rm iso} = 4\pi D^2 F$ obtains $L_{\rm iso} = 2\cos i\, L$, which overestimates the true luminosity by up to a factor of two for face-on disks ($L_{\rm iso}/L = 1.73$ for the AGN at $i = 30°$) and underestimates it for $i > 60°$.

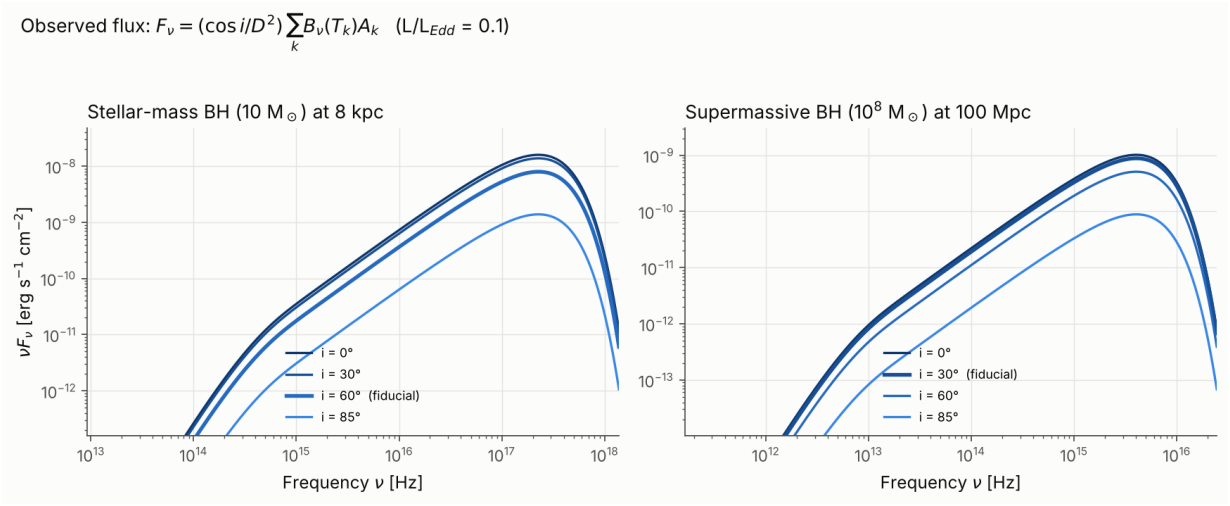


Figure 9: Observed νF_ν for the $10\,M_\odot$ black hole at 8 kpc (left) and the $10^8\,M_\odot$ black hole at 100 Mpc (right), both at $\lambda = 0.1$, for inclinations of 0°, 30°, 60° and 85°. The fiducial inclination is drawn with a thicker line.

5 Comparison with observations

We now test the model against three real datasets (Table 3). Two concern supermassive black holes, through the optical–ultraviolet continua of quasars; the third is a stellar-mass black hole, through X-ray spectra. All data were downloaded from public archives with `fetch_observations.py` and analysed with `compare_observations.py`, both in the `scripts/` directory.

Table 3: Observational data used in this section.

Dataset	Content	Compared with
SDSS composite quasar [14]	median spectrum of 2204 quasars, 800–8555 Å	disk spectrum of a typical SDSS quasar
SDSS DR7 quasar catalogue [15]	virial M, $L/L_{\rm Edd}$ and continuum slopes of $\sim 10^5$ quasars	model slope for each quasar's own M and $L/L_{\rm Edd}$
RXTE/PCA spectra of LMC X-3 (HEASARC)	65 pointings, 1996–1999, 3–20 keV	model folded through the detector response, with M, i, D from Orosz et al. [16]

5.1 The SDSS composite quasar spectrum

Vanden Berk et al. [14] combined 2204 SDSS quasar spectra into a median composite. Measured through their continuum windows at 1350–1365 and 4200–4230 Å, the composite has $f_\nu \propto \nu^{\alpha_\nu}$ with $\alpha_\nu = -0.47$. The paper quotes -0.44 for 1300–5000 Å. Figure 10 compares the composite with the disk spectrum of a typical SDSS DR7 quasar, taken as the sample medians $M = 10^{9.0}\,M_\odot$ and $L/L_{\rm Edd} = 0.17$ (so $T_{\rm max} = 5.1 \times 10^4$ K), normalised at 1350 Å.

For such a disk the spectral peak lies near 1100 Å, so the 1350–4200 Å range is on the rising side of the peak rather than in the $\nu^{1/3}$ regime, and the model gives $\alpha_\nu = -0.24$. The disk shape is qualitatively right: it has the turnover in the far UV that defines the "big blue bump". However, it is bluer than the composite, and it falls well below the composite longward of ~ 4000 Å. There the composite steepens to $\alpha_\nu = -2.45$, which Vanden Berk et al. [14] attribute partly to host-galaxy starlight, a component the disk model does not include.

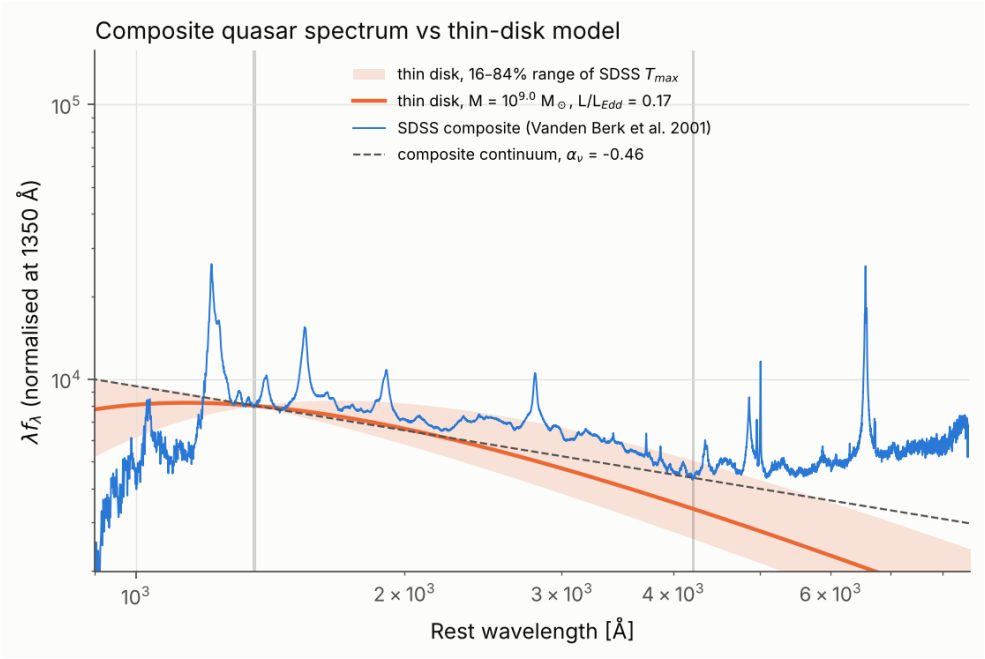


Figure 10: SDSS median composite quasar spectrum [14] (blue) compared with the thin-disk model for the median mass and Eddington ratio of SDSS DR7 quasars (orange line). The shaded band spans the 16–84% range of $T_{\max}$ in that sample. All curves are normalised in the 1350–1365 Å window. Grey bars mark the two continuum windows; the dashed line is the composite's power-law continuum.

5.2 Continuum slopes of individual SDSS quasars

A more direct test uses individual quasars. For every SDSS DR7 quasar, Shen et al. [15] list a virial black hole mass and an Eddington ratio. They also list the slope α_λ of a local power-law continuum $f_\lambda \propto \lambda^{\alpha_\lambda}$, fitted together with an iron template in fixed rest-frame windows around C IV ([1445,1465]+[1700,1705] Å), Mg II ([2200,2700]+[2900,3090] Å) and Hβ ([4435,4700]+[5100,5535] Å). Their Eddington ratios use bolometric corrections from the mean quasar SED of Richards et al. [17].

For each quasar we compute the slope that the disk model predicts in the same windows, given that quasar's M and $L/L_{\rm Edd}$. For fixed $r_{\rm out}/r_g$ the disk spectrum has a universal shape set by $T_{\max}$ alone, with $T_{\max} \propto (\lambda_{\rm Edd}/M)^{1/4}$. The model slope can therefore be tabulated once as a function of $T_{\max}$ and interpolated for each object. We use non-BAL quasars with slope uncertainty < 0.3 and median S/N≥ 5 in the line region.

Table 4: Observed continuum slopes of SDSS DR7 quasars [15] compared with the thin-disk prediction for each quasar's own mass and Eddington ratio. Ranges are 16th–84th percentiles; "obs−model" is the median difference and "redder" the fraction of quasars with a larger (redder) α_λ than predicted.

Window	N	observed α_λ	model α_λ	obs−model	redder
C IV	19 020	−1.30 (−1.74, −0.78)	−1.32 (−1.70, −0.78)	−0.03	48%
Mg II	65 594	−1.36 (−1.71, −0.98)	−1.86 (−2.03, −1.60)	+0.46	86%
Hβ	16 472	−1.51 (−2.03, −0.82)	−2.14 (−2.22, −2.00)	+0.61	88%

The results (Table 4, Figure 11) are mixed. In the far UV (C IV), the observed and predicted distributions agree in both median and width. In the near UV and optical, real quasars are systematically redder than the disk, by $\Delta\alpha_\lambda \approx 0.5$ (Mg II) and 0.6 (Hβ). The predicted Hβ slopes lie close to the $\nu^{1/3}$ limit, $\alpha_\lambda = -7/3$, but the observed ones are much shallower and more widely spread.

The lower panels of Figure 11 show a second difference. The model predicts that the slope

should flatten steeply for cooler disks, but the observed median slope barely changes with the $T_{\max}$ inferred from M and $L/L_{\rm Edd}$. The far-UV agreement in the median is therefore partly a coincidence of where the model curve crosses the data.

Several effects not in the model plausibly contribute:

- dust reddening in the host galaxy;
- host-galaxy starlight in the Hβ windows;
- Balmer-continuum and Fe II emission near Mg II;
- the ~ 0.4 dex uncertainty of single-epoch virial masses, which blurs any trend with $T_{\max}$.

The model may also be incomplete for AGN disks: its outer regions are self-gravitating, and their atmospheres are not blackbodies.

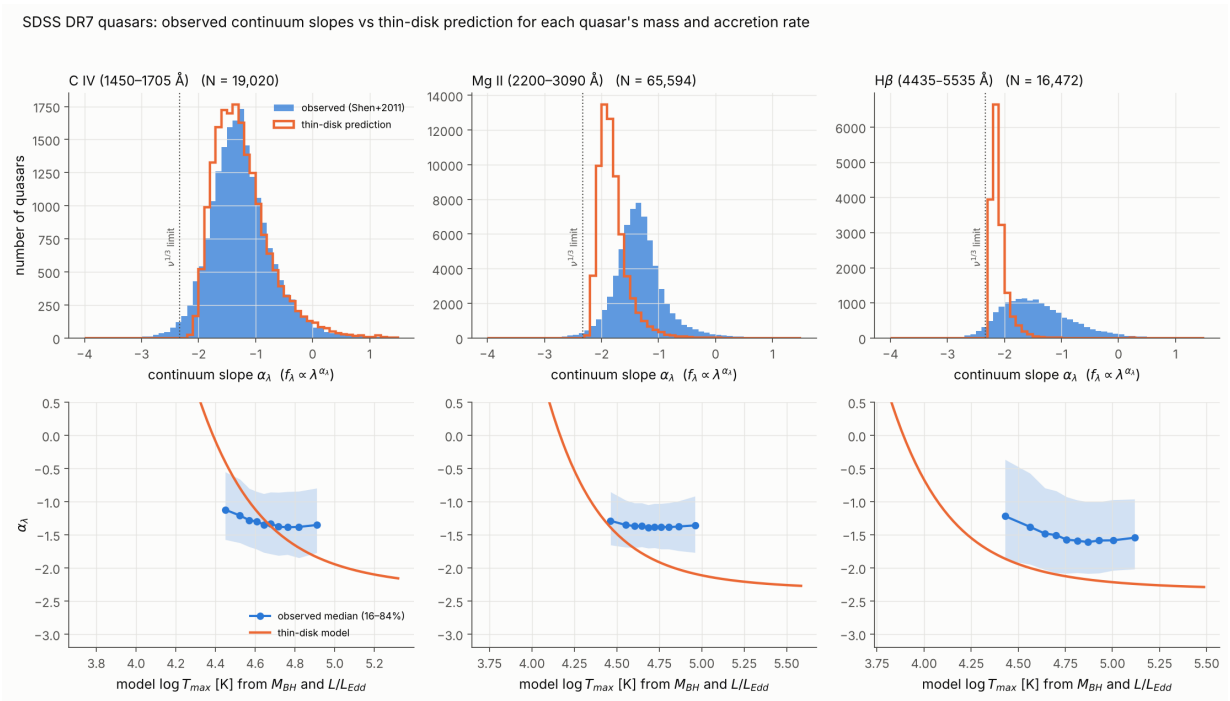


Figure 11: Top: distributions of the observed continuum slope α_λ [15] (filled) and of the thin-disk prediction for the same quasars (outline) in the C IV, Mg II and Hβ windows. The dotted line marks the $\nu^{1/3}$ limit, $\alpha_\lambda = -7/3$. Bottom: observed median slope (points; shading 16–84%) in bins of the model $T_{\max}$ inferred from each quasar's mass and Eddington ratio, compared with the model prediction (line).

5.3 X-ray spectra of the black hole binary LMC X-3

LMC X-3 is a persistent black hole X-ray binary in the Large Magellanic Cloud. It is usually in a disk-dominated (thermal) state and has little interstellar absorption, which makes it a clean test of thin-disk models. Its black hole mass $M = 6.98 \pm 0.56\,M_\odot$ and inclination $i = 69.24 \pm 0.72^\circ$ were measured dynamically by Orosz et al. [16]. We adopt the LMC distance $D = 48.1 \pm 2.2$ kpc used for LMC X-1 by Orosz et al. [18], as Steiner et al. [19] did.

Data. We use the RXTE Proportional Counter Array (PCA) standard-product spectra from the HEASARC archive: all 65 pointings of 1996–1999 with at least 3 ks of exposure. Each pointing comes with a source spectrum, a background spectrum and a response matrix, summed over the PCA detectors (effective area $\approx 6400\,\mathrm{cm}^2$ at 8 keV).

We fit the background-subtracted count rates over 3–20 keV, adding a 1% systematic error per channel as Steiner et al. [19] did. The absorbing column $N_{\rm H} = 4 \times 10^{20}\,\mathrm{cm}^{-2}$ attenuates the flux by less than 1% above 3 keV, so we neglect it.

Model. The ring-sum disk of Section 3 is used with M, i and D fixed to the values above. Each ring radiates a colour-corrected (diluted) blackbody, $f_{\rm col}^{-4} B_\nu(f_{\rm col} T)$. The hardening factor $f_{\rm col}$ raises the colour temperature without changing the emitted flux [20]. A power law represents the weak Comptonised tail.

The model photon spectrum is folded through each response matrix and fitted by least squares. There are four free parameters: $\dot{M}$, $f_{\rm col}$, and the power-law normalisation and index. With $r_{\rm in}$ fixed at $6r_g$, these determine both the colour temperature and the normalisation, $\propto \cos i\, r_{\rm in}^2/(D^2 f_{\rm col}^4)$, so $f_{\rm col}$ is constrained by the data. Following Steiner et al. [19], we call a spectrum thermal-state if the disk provides more than 75% of the 3–20 keV counts and $\chi^2_\nu < 2$. 48 of the 65 spectra qualify.

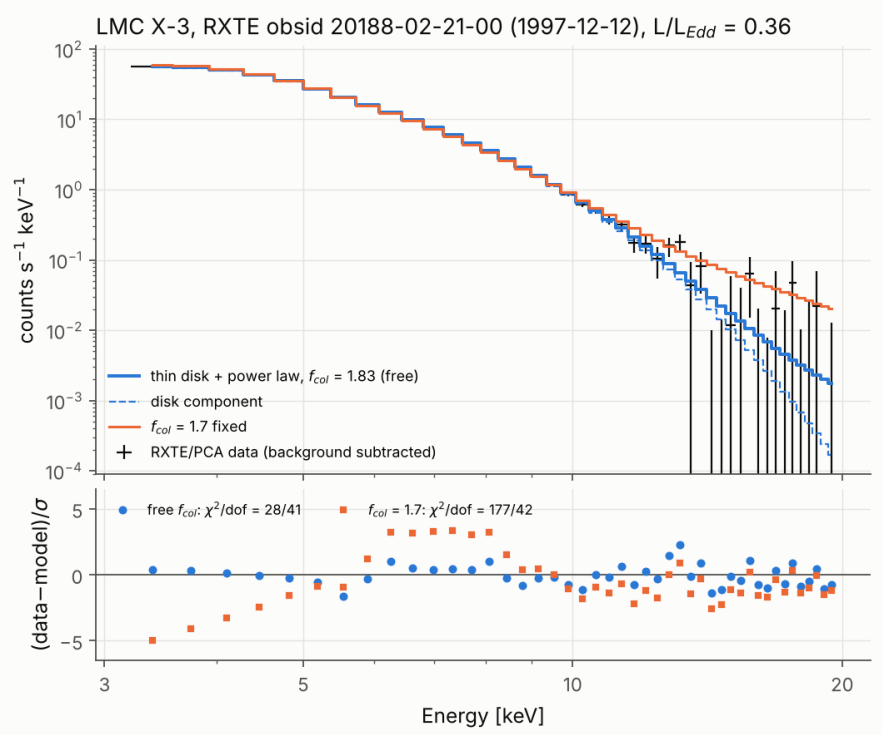


Figure 12: RXTE/PCA spectrum of LMC X-3 (obsid 20188-02-21-00, 1997 December 12) with the best-fitting thin disk plus power law ($f_{\rm col}$ free, blue) and with $f_{\rm col} = 1.7$ fixed (orange). The lower panel shows the residuals.

Results. Figure 12 shows a representative fit. With $f_{\rm col}$ free, the model describes the thermal-state spectra well, with median $\chi^2_\nu = 0.79$. The fitted Eddington ratios span $L/L_{\rm Edd} \approx 0.16$–$0.87$; these are model-dependent, as discussed below. The required hardening factor is remarkably stable from spectrum to spectrum (Figure 13a):

$$f_{\rm col} = 1.81 \;\; (16\text{–}84\%\text{: } 1.78\text{–}1.83) \;\; \pm 0.09 \text{ (syst.)}. \tag{14}$$

The systematic error is obtained by refitting all thermal-state spectra with M, D and i each shifted by $\pm 1\sigma$ (Table 5); the mass dominates. This value agrees with the theoretical $f_{\rm col} \approx 1.7$ of Shimura and Takahara [20].

Fixing $f_{\rm col} = 1.7$ raises the median χ^2_ν to 3.3, mainly because the model's temperature and normalisation cannot then be adjusted independently. Pure blackbody rings ($f_{\rm col} = 1$) are strongly excluded (median $\chi^2_\nu = 117$), which directly demonstrates the spectral hardening discussed in Section 6.

The bolometric disk flux scales as $F \propto T_{\rm col}^{4.4}$ (Figure 13b), close to the $F \propto T^4$ expected if the inner disk edge stays at a fixed radius. This reproduces the result, found with relativistic models, that the inner disk of LMC X-3 sits at a stable radius, identified with the ISCO [19].

Table 5: Median fitted $f_{\rm col}$ of the 48 thermal-state spectra when the adopted system parameters are varied by $\pm 1\sigma$.

Parameter	-1σ	fiducial	$+1\sigma$
M $(6.98 \pm 0.56\,M_\odot)$	1.732	1.806	1.877
D $(48.1 \pm 2.2\,\mathrm{kpc})$	1.848	1.806	1.766
i $(69.24 \pm 0.72^\circ)$	1.820	1.806	1.790

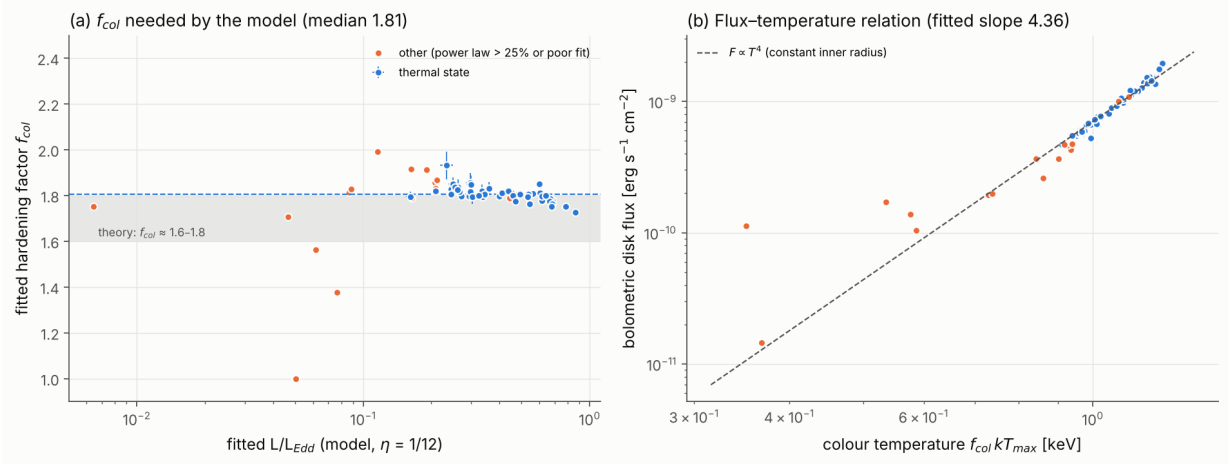


Figure 13: Fits to 65 RXTE/PCA spectra of LMC X-3. (a) Fitted hardening factor versus fitted Eddington ratio. Blue: thermal-state spectra (with 1σ errors); orange: other spectra. The dashed line is the thermal-state median, and the grey band the theoretical range. (b) Bolometric disk flux versus colour temperature $f_{\rm col}kT_{\rm max}$, with the $F \propto T^4$ relation for a constant inner radius.

Caveats. The fitted luminosities depend on the Newtonian efficiency ($\eta = 1/12$) and on the neglect of relativistic effects, which are important at $i \approx 69^\circ$. They should be taken as indicative. For comparison, Steiner et al. [19], using the fully relativistic KERRBB2 model, found that only 391 of their 1454 thermal-state spectra lay in the range $L/L_{\rm Edd} = 0.05$–0.3, so LMC X-3 is frequently more luminous than that. Our $f_{\rm col}$ likewise absorbs any shortcomings of the Newtonian model, as well as the PCA flux calibration (the Crab-based correction of Steiner et al. [19] would change it by ∼2–3%). It is therefore an effective hardening factor rather than a pure measurement of disk-atmosphere physics.

5.4 What the comparisons show

The stellar-mass case is a success. With an independently measured mass, the model reproduces the X-ray spectra of LMC X-3 over a range of luminosities with a single, physically sensible hardening factor. The quasar case is only a partial success. The disk explains the existence and far-UV location of the big blue bump, but real quasar continua are redder in the near-UV and optical than a bare thin disk. Explaining that difference requires components and effects outside the model (Section 6).

6 Discussion

6.1 Connecting observables to accretion physics

The results show how the two basic parameters of the accretion flow map onto observable spectral properties.

- **The bolometric luminosity measures the accretion rate.** Because $L = \eta \dot{M} c^2$ with a mass-independent efficiency, a luminosity measurement (from the flux, distance and inclination) gives $\dot{M}$ directly, but carries no information about M.
- **The spectral peak measures the temperature scale.** The peak frequency gives $T_{\max} \propto (\dot{M}/M^2)^{1/4}$, a combination of mass and accretion rate.
- **Together they determine the mass.** Eliminating $\dot{M}$ between $L = \dot{M}c^2/12$ and Eq. (2) gives

$$M = \left[\frac{3G\dot{M}}{8\pi\sigma T_*^4 \, (6G/c^2)^3} \right]^{1/2} \propto L^{1/2} \, T_{\max}^{-2}. \tag{15}$$

  Applying this inversion to synthetic spectra recovers the input mass to 0.03 % for both the stellar-mass and supermassive cases (see the accompanying notebook). This is the principle behind continuum-fitting estimates of black hole mass and, with relativistic models, spin.

The weak, fourth-root dependence of temperature on both parameters explains why disks around black holes of similar type have similar characteristic energies: stellar-mass disks cluster around 0.5 keV to 2 keV, and AGN disks peak in the ultraviolet. It also means that a large change in accretion rate produces only a modest shift in the spectral peak but a large change in luminosity.

6.2 Limitations

The model captures the essential physics of thermal disk emission, but several effects that matter in comparisons with real data are neglected.

General-relativistic disk structure. The Newtonian temperature profile should be replaced by the relativistic solution of Novikov and Thorne [21]. More importantly, black hole spin moves the ISCO from $6r_g$ (non-rotating) to as close as $1r_g$ for a maximally rotating black hole with a prograde disk [22]. This increases the radiative efficiency from $\approx 5.7\,\%$ (Schwarzschild, compared with our Newtonian $1/12 \approx 8.3\,\%$) to as much as $\approx 42\,\%$, makes the inner disk hotter, and shifts $\nu_{\rm peak}$ upward. Our results therefore correspond most closely to slowly rotating black holes.

Relativistic photon propagation. Near the ISCO the orbital speed is $\sim 0.4c$, so Doppler boosting, gravitational redshift and light bending are significant. They make the observed spectral shape, not only its normalisation, depend on inclination: edge-on disks appear harder and brighter in their high-energy tail than our $\cos i$ scaling predicts.

Departures from blackbody emission. In the hot inner regions of stellar-mass disks electron scattering dominates over absorption, so the emergent spectrum is a diluted, hardened blackbody, usually described by a colour-correction factor $f_{\rm col} \approx 1.7$ [20]. Our fits to LMC X-3 confirm this directly: pure blackbody rings are excluded, and $f_{\rm col} \approx 1.8$ is required (Section 5). Atomic absorption edges and limb darkening also modify the local spectra, particularly in AGN disks.

Additional emission components. Real spectra contain components beyond the thermal disk: a hot, Comptonising corona that produces power-law X-ray emission; reflection features such as the Fe Kα line; synchrotron emission from jets; reprocessing of X-rays in the outer disk (irradiation); and emission from the companion star, the broad-line region, the dusty torus and the host galaxy. Some of these, together with dust reddening, are the likely reason that observed quasar continua are redder than the model in the near-UV and optical (Section 5).

Range of validity of the thin-disk solution. The assumption $H/r \ll 1$ breaks down as L approaches L_{Edd}, where radiation pressure puffs up the inner disk and advection becomes important (slim disks), and at very low accretion rates, where the flow may become radiatively inefficient. AGN disks also become self-gravitating beyond $\sim 10^3$–$10^4 r_g$, so the adopted $r_{\mathrm{out}} = 10^5 r_g$ affects their predicted infrared emission.

Propagation effects. We neglect interstellar and intrinsic absorption, which removes most extreme-ultraviolet photons and hides the peak of AGN disks from direct observation, as well as cosmological redshift and time variability.

7 Conclusions

We have modelled the thermal emission of a geometrically thin, optically thick accretion disk using the standard Shakura–Sunyaev model as implemented in the published `agnpy` package, with `astropy` for the Planck function and physical constants, and by summing the blackbody emission of concentric rings. The ring sum agrees with agnpy's own spectrum to 0.1 % where the latter is valid, converges as N^{-2} and reproduces the analytic luminosity to better than 10^{-5}. Our main findings are:

1. The disk temperature obeys $T_{\max} \propto (\dot{M}/M^2)^{1/4}$, so at a fixed Eddington ratio more massive black holes have cooler disks, $T_{\max} \propto M^{-1/4}$.
2. The same model puts the thermal peak of a $10\,M_\odot$ black hole at $\sim 1\,\mathrm{keV}$ (soft X-rays) and that of a $10^8\,M_\odot$ black hole at $\sim 740\,\text{Å}$ (far UV), a factor of 56 apart in frequency.
3. The bolometric luminosity, $L = \dot{M}c^2/12$, depends only on the accretion rate; the spectral peak, $h\nu_{\mathrm{peak}} \approx 2.47\,kT_{\max}$, moves only as $\dot{M}^{1/4}$.
4. The luminosity and the peak frequency together determine both $\dot{M}$ and M.
5. For an observer, the flux scales as $\cos i/D^2$, and assuming isotropic emission biases the inferred luminosity by a factor of $2\cos i$.
6. Compared with real quasars, the disk matches the average far-UV continuum slope of SDSS quasars. In the near-UV and optical, 86–88% of quasars are redder than the disk predicted for their own mass and Eddington ratio, by $\Delta\alpha_\lambda \approx 0.5$–$0.6$.
7. Fitted to 48 thermal-state RXTE spectra of LMC X-3 with its dynamically measured mass, the model gives good fits with a nearly constant hardening factor $f_{\mathrm{col}} = 1.81 \pm 0.09$. This is consistent with theory, and pure blackbody emission is excluded.

The most important extensions of this work would be to include the relativistic Novikov–Thorne temperature profile with black hole spin, relativistic ray tracing, a colour-correction factor, and a Comptonised coronal component.

A Code and reproducibility

All results in this report can be reproduced from the project repository:

`accretion_disk/disk.py`	`ThinDisk` (agnpy `SSDisk` + ring sum)
`scripts/run_analysis.py`	generates all figures and `results/summary.csv`
`tests/`	validation tests (`python -m pytest`)
`notebooks/accretion_disk_walkthrough.ipynb`	step-by-step walkthrough
`scripts/fetch_observations.py`	downloads the observational data
`scripts/compare_observations.py`	comparisons of Section 5
`data/observations/`	the observational data and their provenance

The code requires Python 3 with `numpy`, `scipy`, `matplotlib`, `astropy` and `agnpy` (see `requirements.txt`). A minimal example:

```
from accretion_disk import ThinDisk, constants as C
disk = ThinDisk.from_eddington(mass_msun=10, eddington_ratio=0.1)
nu   = disk.frequency_grid()
L_nu = disk.luminosity_nu(nu)                                  # erg/s/Hz
F_nu = disk.flux_nu(nu, distance=8*C.kpc, inclination_deg=60)  # erg/s/cm^2/Hz
print(disk.T_max, disk.peak_frequency(), disk.bolometric_luminosity())
print(disk.ssdisk)      # the underlying agnpy.targets.SSDisk object
```

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